

FSD Laboratory 04

Name : Nikhil Karmare

Roll No : 38

PRN : 1262243774

Panel : F

Aim: Write server-side script in PHP to perform form validation and create database application using PHP and MySQL to perform insert, update, delete and search operations.

Objectives:

1. To understand Server-side Scripting.
2. To learn database connectivity using PHP-MySQL.
3. To perform insert, update, delete and search operations on database.

Theory:

1. PHP Architecture.
⇒ PHP follows a **client-server architecture**:
 - Client (Browser) sends request
 - Web server (Apache) receives request
 - PHP engine processes .php file
 - Database interaction (MySQL) happens
 - Output (HTML) sent back to browser

Flow:

Browser → Web Server → PHP → Database → PHP → Browser

2. Steps for Database connectivity in PHP.
⇒ ☐ Create Database
☐ Create Table
☐ Connect PHP to MySQL (MySQLi / PDO)
☐ Execute Query
☐ Fetch Data
☐ Close Connection

FAQ:

1. What are the advantages of Server-side Scripting?
⇒
 - Secure (code hidden from user)
 - Handles database operations
 - Dynamic content generation
 - Authentication & session handling

2. What is XAMPP and phpMyAdmin?

- ⇒ ☐ **XAMPP**: Local server package (Apache + MySQL + PHP)
☐ **phpMyAdmin**: GUI tool to manage MySQL databases\

3. What are the two ways to connect to a database in PHP?

- MySQLi
- PDO (PHP Data Objects)

Output: Screenshots of the output to be attached.

Sample Problem Statements:

PHP CRUD Operations

1. Student can create a PHP form or use existing/ implemented HTML form for Student's Registration System with the fields mentioned: First name, Last name, Roll No/ID, Password, Confirm Password, Contact number and perform following operations

1. Insert student details - First name, Last name, Roll No/ID, Password, Confirm Password, Contact number
2. Delete the Student records based on Roll no/ID
3. Update the Student details based on Roll no/ID- Example students can update their contact details based on searching the record with Roll no.
4. Display the Updated student details or View the Students record in tabular format.

Apply Form Validation on the necessary fields using PHP/Javascript

2. Student can create a PHP form or use existing/ implemented HTML form for Library Management System with the fields mentioned: Book name, ISBN No, Book title, Author name, Publisher name and perform following operations

1. Insert Book details - Book name, ISBN No, Book title, Author name, Publisher name
2. Delete the Book records based on ISBN No
3. Update the Book details based on ISBN No- Example students can update wrong entered book details based on searching the record with ISBN No.
4. Display the Updated Book details or View the Book Details records in tabular format.

Apply Form Validation on the necessary fields using PHP/Javascript

3. Student can create a PHP form or use existing/ implemented HTML form for Employee Management System with the fields mentioned: Employee name, Employee ID, Department_name, Phone number, Joining Date and perform following operations

1. Insert Employee details - Employee name, Employee ID, Department_name, Phone number, Joining Date
2. Delete the Employee records based on Employee ID

3. Update the Employee details based on Employee ID- Example students can update Employee details based on searching the record with Employee ID.

4. Display the Updated Employee details or View the Employee Details records in tabular format.

Apply Form Validation on the necessary fields using PHP/Javascript

4. Student can create a PHP form or use existing/ implemented HTML form for Flight Booking Management System with the fields mentioned: Passenger name, From, to, date, Departure date, Arrival date, Phone number, Email ID and perform following operations

1. Insert Passenger details - Passenger name, From, to, date, Departure date, Arrival date, Phone number, Email ID

2. Delete the Passenger records based on Phone Number

3. Update the Passenger details based on Phone Number - Example students can update Flight Booking details based on searching the record with Phone Number.

4. Display the Updated Flight Booking details or View the Flight Booking Details records in tabular format.

Apply Form Validation on the necessary fields using PHP/Javascript.

Technologies Student Should Use:

XAMPP

PHP for Server-side Scripting

MySQL as a backend Database

Student Registration

First Name: Nikhil

Last Name: Karmare

Roll No: 38

Password:

Confirm Password:

Contact Number: 7249512929

Register

[View Students](#)

Student Records

Roll No	First Name	Last Name	Contact	Delete	Update
38	Nikhil	Karmare	7249512929	Delete	Update
56	ggg	grg	4344343rr4	Delete	Update

```
Restarting XAMPP for Mac OS X 8.2.4-0...
XAMPP: Stopping Apache...ok.
XAMPP: Stopping MySQL...not running.
XAMPP: Stopping ProFTPD...ok.
XAMPP: Starting Apache...ok.
XAMPP: Starting MySQL...ok.
XAMPP: Starting ProFTPD...ok.
nikhil@nikhil2113 ~ % sudo /Applications/XAMPP/xamppfiles/xampp startmysql
Password:
XAMPP: Starting MySQL...ok.
nikhil@nikhil2113 ~ % cd /Applications/XAMPP/htdocs/student_crud
nikhil@nikhil2113 student_crud % git init
Initialized empty Git repository in /Applications/XAMPP/xamppfiles/htdocs/student_crud/.git/
nikhil@nikhil2113 student_crud % cd ~/Documents/nikhil_FSDL
nikhil@nikhil2113 nikhil_FSDL % git checkout main
git pull origin main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
From https://github.com/nikhilkarmare/nikhil_FSDL
* branch      main      -> FETCH_HEAD
Already up to date.
nikhil@nikhil2113 nikhil_FSDL % git checkout -b assignment4
Switched to a new branch 'assignment4'
nikhil@nikhil2113 nikhil_FSDL % git branch
  assignment2
  assignment3
* assignment4
  main
nikhil@nikhil2113 nikhil_FSDL % mv "/Applications/XAMPP/htdocs/student_crud" .
nikhil@nikhil2113 nikhil_FSDL % ls
file1.txt      file2.txt      file3.txt      LICENSE      student_crud
nikhil@nikhil2113 nikhil_FSDL % git add student_crud
error: 'student_crud/' does not have a commit checked out
fatal: adding files failed
nikhil@nikhil2113 nikhil_FSDL % cd student_crud
nikhil@nikhil2113 student_crud % rm -rf .git
nikhil@nikhil2113 student_crud % cd ..
git add student_crud
nikhil@nikhil2113 nikhil_FSDL % git commit -m "Assignment 4: Added student CRUD PHP project"
[assignment4 05eb9cb] Assignment 4: Added student CRUD PHP project
 7 files changed, 184 insertions(+)
```

```
create mode 100644 student_crud/db.php
create mode 100644 student_crud/delete.php
create mode 100644 student_crud/index.php
create mode 100644 student_crud/insert.php
create mode 100644 student_crud/style.css
create mode 100644 student_crud/update.php
create mode 100644 student_crud/view.php
nikhil@nikhil2113 nikhil_FSDL % git push origin assignment4
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 14 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 2.03 KiB | 2.03 MiB/s, done.
Total 10 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'assignment4' on GitHub by visiting:
remote:   https://github.com/nikhilkarmare/nikhil_FSDL/pull/new/assignment4
remote:
To https://github.com/nikhilkarmare/nikhil_FSDL.git
* [new branch]      assignment4 -> assignment4
nikhil@nikhil2113 nikhil_FSDL % git checkout assignment4
Already on 'assignment4'
nikhil@nikhil2113 nikhil_FSDL % git rm file1.txt file2.txt file3.txt LICENSE
rm 'LICENSE'
rm 'file1.txt'
rm 'file2.txt'
rm 'file3.txt'
nikhil@nikhil2113 nikhil_FSDL % git commit -m "Assignment 4: Removed unnecessary files from branch"
[assignment4 453dc77] Assignment 4: Removed unnecessary files from branch
 4 files changed, 204 deletions(-)
delete mode 100644 LICENSE
delete mode 100644 file1.txt
delete mode 100644 file2.txt
delete mode 100644 file3.txt
nikhil@nikhil2113 nikhil_FSDL % git push origin assignment4
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 14 threads
Compressing objects: 100% (1/1), done.
Writing objects: 100% (2/2), 267 bytes | 267.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/nikhilkarmare/nikhil_FSDL.git
 05eb9cb..453dc77  assignment4 -> assignment4
nikhil@nikhil2113 nikhil_FSDL %
```